Table 1: Signed change in metabolic tumor volume (MTV) and total lesion glycolysis (TLG) caused by registration on the test set (26 patients, 30 lesion-positive pairs): positive values mean that the warped lesions are larger or have greater total uptake. Methods are as in Table **??**; Training / IO indicates where losses are used, and PET + rig. denotes PET-preservation and rigidity losses. Values are mean±SD and median over patients after averaging each patient's eligible pairs.

			ΔMTV (%)		ΔTLG (%)	
Method	Training	IO	Mean±SD	Median	Mean±SD	Median
Affine	–	–	$+2.9 \pm 5.8$	$+1.3$	$+2.8 \pm 5.7$	$+1.7$
NiftyReg	–	–	$+4.7 \pm 18.9$	-0.6	$+7.1 \pm 23.8$	-0.1
ConvexAdam	–	–	$+8.0 \pm 8.1$	$+7.1$	$+8.8 \pm 10.3$	$+5.8$
Affine + IO	–	PET + rig.	$+3.5 \pm 5.8$	$+3.8$	$+5.7 \pm 5.6$	$+6.7$
Ours	–	–	$+5.3 \pm 13.1$	$+4.5$	$+7.9 \pm 14.9$	$+7.0$
Ours	PET + rig.	–	$+5.1 \pm 11.0$	$+3.4$	$+7.6 \pm 12.5$	$+5.2$
Ours + IO	–	basic	$+1.1 \pm 16.5$	$+2.1$	$+4.0 \pm 20.0$	$+3.6$
Ours + IO	PET + rig.	PET + rig.	$+4.2 \pm 11.8$	$+3.0$	$+7.3 \pm 12.8$	$+6.4$
Ours + IO	–	PET + rig.	$+3.2 \pm 13.5$	$+3.1$	$+6.2 \pm 14.8$	$+6.2$